Universality of Event Density Architecture:
Structural Invariants Across Parameters, Conditions, and Dimensions

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Abstract
Event Density (ED) exhibits a reproducible architectural organization—saddles, channels, boundaries, horizons, and coherent regimes—that persists across a wide range of parameters, initial conditions, physical measures, and dimensions. This paper demonstrates that these motifs are not contingent features of specific simulations but universal consequences of the ED rule. Through systematic variation of mobility laws, noise amplitude, suppression depth, and nonlinear amplification, we show that the architectural motifs remain identifiable and stable. Across diverse initial conditions—random, smooth, structured, and adversarial—the system converges toward the same architectural attractors. Extending the framework to three dimensions reveals natural geometric generalizations: saddles become saddle-sheets, channels become tubes, boundaries become walls, and horizons become surfaces. Physical quantities introduced in ED-Arch-03—gradient energy, mobility energy, entropy-like measures, and flux—exhibit the same invariants, reinforcing the architectural layer with independent physical signatures. The four canonical regimes appear in every parameter family and dimension, and their transitions occur in a consistent order. Together, these results establish ED as a general theory of architectural emergence: a framework in which coherent structure arises inevitably from the interplay of gradients, curvature, mobility, and noise, independent of how the system begins or how it is perturbed.

1. Introduction — Universality as the Signature of Event Density
Event Density (ED) exhibits a striking and repeatable architectural organization: saddles, channels, boundaries, horizons, and coherent regimes arise across a wide range of parameters, initial conditions, and dimensions. The previous papers in this series established the empirical reality of this architecture (ED-Arch-01), formalized its ontological structure (ED-Arch-02), and derived its physical interpretation in terms of energy, entropy, and flux (ED-Arch-03). The present paper advances the framework by addressing a deeper question: Are these architectural motifs universal?

Universality is the hallmark of a genuine scientific ontology. A structure is universal when it persists under deformation—when it survives changes in parameters, noise, mobility laws, dimensionality, and initial conditions. In statistical mechanics, universality explains why disparate systems share the same critical exponents. In dynamical systems, it explains why bifurcations recur across models. In ED, universality would mean that the architectural motifs are not artifacts of a particular simulation or rule choice, but necessary consequences of the ED update itself.

This paper demonstrates that ED's architecture is universal in precisely this sense. Across extensive parameter sweeps, diverse initial conditions, and dimensional uplift from 2D to 3D, the same structural invariants reappear: gradient sign-flip loci that generate saddles; mixed-curvature corridors that stabilize channels; positive-curvature regions that form boundaries; negative-curvature regions that form horizons; and the four canonical regimes that organize global behavior. These motifs persist even when mobility laws are altered, suppression depth is varied, or noise is increased up to the critical threshold. The architecture is not imposed—it is inevitable.

The goal of this paper is to establish this inevitability. We show that the architectural motifs identified in earlier papers arise under all tested conditions, that the regime transitions occur in the same order across parameter

families, and that the curvature-driven logic of ED generalizes naturally to higher dimensions. The result is a universality thesis: the ED architecture is intrinsic to the rule, independent of how the system begins, how it is perturbed, or how many dimensions it occupies.

The paper proceeds by demonstrating universality under parameter variation, initial conditions, dimensional uplift, and physical measures, culminating in a synthesis that positions ED as a general theory of architectural emergence.

This paper contributes: (1) a demonstration of parameter universality, (2) initial-condition universality, (3) dimensional universality, (4) physical universality, (5) regime universality, and (6) a synthesis establishing ED as a general architectural framework.

Methods
All simulations were performed using the ED-Simulation engine introduced in ED-Arch-01 and extended in ED-Arch-03. The system evolves a scalar event-density field $E(x,t)$ on a discrete lattice using a local update rule that combines gradient-driven mobility, curvature-driven reorganization, stochastic noise, and suppression. Parameter sweeps were conducted by varying mobility exponent, noise amplitude, suppression depth, and nonlinear amplification across broad ranges, while initial conditions included random, smooth, structured, and adversarial configurations. Dimensional uplift was implemented by extending the same update rule to a 3D lattice, preserving gradient, curvature, and mobility operators in their natural higher-dimensional forms.

2. Universality Under Parameter Variation

The first axis of universality concerns the parameters that govern the ED update rule. If the architectural motifs of Event Density are intrinsic to the rule, then they must persist across wide ranges of mobility, noise, suppression depth, and nonlinearity. This section demonstrates precisely that: the ED architecture is stable under parameter deformation, and the motifs identified in earlier papers—saddles, channels, boundaries, horizons, and regimes—reappear consistently across the parameter space.

Parameter variation is not merely a robustness test; it is a structural probe. Each parameter influences a different geometric mechanism of ED: mobility sharpens gradients, noise perturbs curvature, suppression shapes wells and pockets, and nonlinear amplification modulates the strength of curvature-driven reorganization. If the architecture persists despite these perturbations, then its origin must lie deeper than any particular parameter choice. The results presented here show that this is indeed the case.

2.1 Mobility Exponent and Structural Sharpness
The mobility exponent α controls how strongly the system responds to gradients. High mobility amplifies coherent structure, sharpening channels and stabilizing saddles; low mobility smooths gradients and weakens boundaries. Yet across all tested values of α, the same architectural motifs appear. Saddles form wherever gradient sign-flips occur, channels align along mixed-curvature corridors, and boundaries and horizons emerge from curvature transitions. Mobility affects the sharpness of these structures, not their existence. This demonstrates that the architectural logic is independent of the specific mobility law.

2.2 Noise Amplitude and the Critical Threshold

Noise amplitude σ determines whether curvature can organize into coherent motifs or is overwhelmed by stochastic fluctuations. As shown in previous papers of this series, there exists a critical noise threshold σ_c above which the system enters the non-individuating regime. The universality result is that this threshold—and the

sequence of regime transitions it induces—is consistent across all parameter families. Increasing noise always drives the system through the same progression:

hyper-coherent → stable-coherent → marginal → non-individuating

This ordering is invariant under changes in mobility, suppression depth, and nonlinearity. Noise perturbs the architecture, but it does not alter its fundamental structure.

2.3 Suppression Depth and Structural Wells

Suppression depth β shapes the formation of wells, pockets, and persistent cores. Deep suppression stabilizes coherent structures; shallow suppression allows them to drift or dissolve. Yet across all tested values of β, the same architectural motifs appear. Saddles remain identifiable, channels remain aligned, and boundaries and horizons continue to form wherever curvature transitions occur. Suppression depth modulates the longevity of structures, not the architecture itself. This demonstrates that ED's motifs are not artifacts of a particular balance between accumulation and dissipation.

2.4 Gradient Amplification and Nonlinearity

Nonlinearity γ influences how curvature reorganizes under the ED update. Strong nonlinear amplification sharpens curvature transitions, while weak amplification smooths them. Yet the determinant-zero surfaces of the Hessian—the architectural hinges of ED—persist across all tested nonlinearities. Mixed curvature still produces saddles, positive curvature still forms boundaries, and negative curvature still forms horizons. The architecture is therefore not tied to a specific functional form of curvature response; it arises from the geometric structure of the ED rule itself.

2.5 Synthesis: Parameter-Invariant Architecture

Across all parameter variations, the same conclusion emerges: the ED architecture is stable under deformation. Saddles, channels, boundaries, horizons, and regime transitions appear consistently across mobility laws, noise levels, suppression depths, and nonlinearities. The motifs sharpen or soften, persist longer or shorter, or shift slightly in position, but they do not disappear. This invariance shows that the architectural motifs are not contingent features of a particular parameter choice—they are structural consequences of the ED update rule.

3. Universality Across Initial Conditions

If the architectural motifs of Event Density are intrinsic to the rule, then they must emerge regardless of how the system begins. Initial conditions are the strongest test of universality because they probe whether the architecture is imposed by the starting configuration or generated by the dynamics themselves. A genuine architectural ontology cannot depend on a privileged initial state; it must arise from the rule under all admissible beginnings. This section demonstrates that ED satisfies this criterion: the same motifs—saddles, channels, boundaries, horizons, and regime transitions—appear across all tested initial conditions, from smooth fields to adversarial configurations.

The ED update amplifies gradient sign-flips, sharpens curvature transitions, and suppresses unstructured fluctuations. These mechanisms operate independently of the initial state, driving the system toward the same architectural attractors. The result is a convergence phenomenon: different beginnings lead to the same structural motifs, not because the system forgets its past, but because the ED rule actively reorganizes it.

3.1 Random Noise

Random noise provides the most unstructured initial condition. Despite the absence of coherent gradients or curvature, the ED dynamics rapidly generate sign-flip loci, which seed saddles. Curvature transitions emerge as the Hessian organizes, producing boundaries and horizons. Channels form as gradients align along mixed-curvature corridors. The resulting architecture is indistinguishable from that produced by smooth or structured initial conditions. This demonstrates that ED does not require pre-existing structure; it generates structure from stochastic beginnings.

3.2 Smooth Fields

Smooth initial fields contain minimal curvature and few gradient sign-flips. Yet even in this case, the ED update introduces curvature through mobility-driven sharpening and suppression-driven accumulation. Small perturbations grow into saddles, channels, and boundaries. The architecture emerges more slowly than in noisy conditions, but the end state is the same: a coherent network of saddles and channels bounded by curvature-defined surfaces. Smoothness delays architecture but does not prevent it.

3.3 Structured Initial Conditions: Stripes, Bubbles, Fractal Seeds

Structured initial conditions test whether ED merely preserves imposed geometry or reorganizes it. In all cases, the imposed structure is transient. Stripes break into saddle-aligned segments; bubbles collapse into boundary-defined pockets; fractal seeds reorganize into coherent channels and horizons. The ED rule does not preserve arbitrary structure; it replaces it with the architectural motifs dictated by curvature and gradient dynamics. This demonstrates that ED is not a pattern-preserving system but a pattern-organizing one.

3.4 Adversarial Initial Conditions

Adversarial initial conditions—configurations designed to frustrate saddle formation or channel alignment—provide the strongest test of universality. Even under these conditions, the ED update generates curvature transitions and gradient sign-flips that reorganize the field into the canonical motifs. Saddles appear in regions where the adversarial configuration cannot suppress sign-flip formation; channels emerge along mixed-curvature corridors; boundaries and horizons form wherever curvature transitions occur. Adversarial structure delays architecture but cannot prevent it.

3.5 Convergence as an Architectural Invariant

Across all initial conditions, the same conclusion emerges: the ED architecture is independent of how the system begins. Random noise, smooth fields, structured patterns, and adversarial configurations all converge toward the same set of motifs. This convergence is not accidental; it is driven by the geometric mechanisms of the ED rule:

- gradient sign-flip → saddle formation
- curvature transitions → boundaries and horizons
- coherent gradient alignment → channels
- noise and mobility → regime transitions

These mechanisms operate universally, ensuring that the architecture arises from the rule itself rather than from the initial state.

4. Universality Across Dimensionality

A defining test of any architectural framework is whether its structural motifs persist when the dimensionality of the system changes. Many models exhibit coherent organization in two dimensions but lose structure when extended to three; others rely on planar geometry in ways that do not generalize. Event Density is different. The

architectural motifs identified in earlier papers—saddles, channels, boundaries, horizons, and coherent regimes—reappear in higher dimensions with natural geometric generalizations. This section demonstrates that ED's architecture is dimension-agnostic: the same curvature transitions, gradient alignments, and regime structures persist when the system is extended from 2D to 3D.

Dimensional uplift is not merely a robustness test; it is a structural probe. If the ED architecture were an artifact of planar geometry, it would collapse or transform unpredictably in 3D. Instead, the motifs become richer: points become surfaces, curves become tubes, and boundaries and horizons become volumetric manifolds. The underlying logic—gradient sign-flip, curvature transitions, and mobility-driven sharpening—remains unchanged. This invariance shows that ED's architecture is not tied to the specifics of 2D geometry but arises from deeper principles that scale with dimension.

4.1 Saddles Become Saddle-Sheets

In 2D, a saddle is defined by one compressive and one expansive principal direction. In 3D, the Hessian has three eigenvalues, and the saddle generalizes to a saddle-sheet: a 2D surface where one eigenvalue has opposite sign from the other two. This surface partitions space into regions of compression and expansion, just as the saddle point does in 2D. Channels and boundaries attach to the saddle-sheet along its principal curvature directions, forming the organizing scaffold of 3D ED architecture. The transition from saddle to saddle-sheet is smooth and mathematically natural, demonstrating that the core architectural unit persists under dimensional uplift.

4.2 Channels Become Tubes

Channels in 2D are narrow corridors of aligned gradient flow. In 3D, these become tubes: elongated, coherent structures where the gradient maintains alignment along a spatial curve embedded in three-dimensional space. Tubes exhibit confining curvature in two transverse directions and mixed curvature along the longitudinal direction, mirroring the channel's structure in 2D. Tubes can twist, branch, and merge in ways impossible in planar geometry, enriching the architectural possibilities without altering the underlying logic. Tubes connect saddle-sheets into networks, forming the backbone of coherent 3D regimes.

4.3 Boundaries Become Walls

Boundaries in 2D are compression-dominated curves where gradient vectors converge. In 3D, these generalize to walls: 2D surfaces where curvature is positive in both transverse directions. Walls enclose coherent volumes, regulate the stability of tubes, and form the architectural "containers" of 3D ED. Their formation is driven by the same curvature transitions that produce boundaries in 2D, showing that the architectural logic persists unchanged.

4.4 Horizons Become Surfaces

Horizons in 2D are expansion-dominated curves where gradient vectors diverge. In 3D, they become surfaces: 2D manifolds where curvature is negative in both transverse directions. Horizons mark transitions from structured interiors to unstructured exteriors, regulate the flow of noise, and determine whether coherent volumes persist or dissolve. Their geometry is richer in 3D but arises from the same determinant-zero transitions that define horizons in 2D.

4.5 Regimes Become Volumetric Phases

The four canonical regimes—hyper-coherent, stable-coherent, marginal, and non-individuating—reappear in 3D with volumetric expression. Hyper-coherent regimes contain stable networks of tubes and saddle-sheets; stable-coherent regimes maintain persistent but deformable volumetric structures; marginal regimes exhibit flickering tubes and transient sheets; and non-individuating regimes collapse into noise-dominated churn. The

phase diagram is unchanged; only the geometry becomes richer. This demonstrates that the regime structure is a universal feature of the ED rule, not a property of planar dynamics.

4.6 Curvature Transitions Generalize Naturally

The determinant-zero condition that marks curvature transitions in 2D becomes a codimension-1 surface in 3D. This surface separates tube interiors, boundary walls, horizon surfaces, and saddle-sheet regions. The same architectural logic—mixed curvature inside, positive curvature confining, negative curvature expanding—applies in any dimension. This natural generalization shows that the ED architecture is rooted in geometric principles that scale with dimension.

4.7 Dimensional Universality as Architectural Evidence

Across all dimensional extensions, the same conclusion emerges: the ED architecture persists under dimensional uplift. Saddles become saddle-sheets, channels become tubes, boundaries become walls, and horizons become surfaces. The regime structure remains intact, and the curvature-driven logic of ED applies without modification. This dimensional universality is strong evidence that ED's motifs are not artifacts of 2D geometry but necessary consequences of the ED rule.

5. Universality Across Energy and Flux Conditions

The physical interpretation of Event Density introduced in ED-Arch-03—gradient energy, mobility energy, entropy-like measures, and flux across curvature-defined surfaces—provides a second axis along which universality can be tested. If the ED architecture is intrinsic to the rule, then the physical quantities derived from the field should exhibit the same invariants as the geometric motifs. This section demonstrates that they do: the physical layer reinforces the architectural layer, and the same patterns of concentration, flow, and dissipation appear across all parameter families, initial conditions, and dimensions.

Energy and flux are not external additions to ED; they are emergent consequences of the same gradient and curvature dynamics that generate saddles, channels, boundaries, and horizons. The universality of these physical quantities therefore provides independent confirmation that the architecture is not contingent. The physical layer and the geometric layer converge on the same structural motifs, revealing ED as a unified architectural-physical framework.

5.1 Gradient Energy Concentration Around Saddles

Gradient energy measures the local intensity of directional change in the event density field. Across all tested conditions, gradient energy consistently concentrates around saddles. This occurs because saddles arise from gradient sign-flip, which necessarily produces high-magnitude gradients in orthogonal directions. The result is a universal pattern: saddles are always energy-dense loci, regardless of mobility law, noise amplitude, suppression depth, or initial condition. This concentration is a physical signature of the architectural role saddles play as organizing centers.

5.2 Mobility Energy and Channel Stability

Mobility energy quantifies the work performed by mobility-driven flow. Channels, defined by coherent gradient alignment, universally exhibit elevated mobility energy along their length. This pattern persists across all parameter variations: channels are always energy-bearing conduits that transmit individuating flow between saddles. Even when channels soften or drift under parameter deformation, their mobility-energy signature remains intact. This demonstrates that channels are not merely geometric corridors but physically meaningful pathways of flow.

5.3 Entropy-Like Measures and Regime Structure

Entropy-like measures introduced in ED-Arch-03 quantify the degree of disorder in the field. Across all conditions, entropy minima correspond to coherent regimes, while entropy maxima correspond to non-individuating churn. The four canonical regimes—hyper-coherent, stable-coherent, marginal, and non-individuating—reappear in the entropy landscape with the same ordering observed in the geometric analysis. This alignment shows that regimes are not only geometric phases but thermodynamic phases, reinforcing the universality of the ED phase diagram.

5.4 Flux Across Boundaries and Horizons

Flux measures the net flow of event density across curvature-defined surfaces. Boundaries universally exhibit inward flux, reflecting their role as compression-dominated walls that confine structure. Horizons universally exhibit outward flux, reflecting their role as expansion-dominated surfaces that regulate dissipation. These flux patterns persist across all parameter families and initial conditions, demonstrating that boundaries and horizons are physically distinct surfaces, not merely geometric artifacts. The flux signatures confirm the architectural complementarity identified in ED-Arch-02.

5.5 Physical Invariants as Independent Confirmation

The universality of gradient energy, mobility energy, entropy, and flux provides independent confirmation of the architectural invariants. The physical layer and the geometric layer converge on the same motifs:

- Saddles → energy concentration
- Channels → mobility-energy conduits
- Boundaries → inward-flux surfaces
- Horizons → outward-flux surfaces
- Regimes → entropy-defined phases

This convergence is not accidental. It shows that the ED architecture is not merely a geometric phenomenon but a physical necessity of the ED rule. The physical quantities do not introduce new motifs; they reveal the physical meaning of the motifs already present in the architecture.

5.6 Synthesis: Physical Universality as Architectural Evidence

Across all physical measures, the same conclusion emerges: the ED architecture is physically universal. Energy concentrates around saddles, flows along channels, accumulates within boundaries, dissipates across horizons, and organizes into regime-defined phases. These patterns persist under parameter variation, initial condition variation, and dimensional uplift. The physical layer therefore reinforces the architectural layer, demonstrating that ED's motifs are not contingent features of a particular simulation but necessary consequences of the ED rule.

6. Regime Universality

The regime taxonomy introduced and quantified in previous papers of this series—hyper-coherent, stable-coherent, marginal, and non-individuating—captures the global organizational states of the Event Density field. These regimes are not arbitrary labels but structural phases defined by the stability of saddles, the coherence of channels, the persistence of boundaries and horizons, and the ability of curvature to organize against noise. In this section, we show that the regime structure itself is universal: the same four regimes appear across all parameter families, initial conditions, and dimensions, and the transitions between them occur in the same order. This universality establishes the ED phase diagram as a fundamental architectural invariant.

Regimes are the macroscopic expression of ED's microscopic geometry. Saddles, channels, boundaries, and horizons are the local motifs; regimes are the global states that emerge from their interactions. If the motifs are universal, the regimes must be as well. The results presented here confirm this: the ED phase diagram is not contingent—it is intrinsic to the ED rule.

6.1 Hyper-Coherent Regime: Full Architectural Articulation

Across all tested conditions, the hyper-coherent regime appears whenever noise is low, mobility is sufficiently strong, and curvature can fully organize. This regime is characterized by:

- persistent saddles with stable eigenvalue structure
- long, sharply defined channels
- thick, confining boundaries
- well-formed horizons
- coherent networks of individuating flow

The hyper-coherent regime is the fully articulated expression of ED architecture. Its appearance across all parameter families and initial conditions demonstrates that ED possesses a universal high-order phase in which all architectural invariants are present and stable.

6.2 Stable-Coherent Regime: Deformable but Persistent Structure

The stable-coherent regime appears when noise increases or mobility decreases but remains within the subcritical range. Across all conditions, this regime exhibits:

- saddles that persist but may drift
- channels that remain aligned but soften
- boundaries that thin or deform
- horizons that fluctuate at their edges

Despite these deformations, the architectural motifs remain intact. This regime is universal because it arises whenever curvature can organize but cannot fully suppress noise. It is the "working phase" of ED: coherent, dynamic, and robust.

6.3 Marginal Regime: Flickering Proto-Architecture

The marginal regime appears consistently as the transitional phase between coherence and non-individuation. It is characterized by:

- saddles that form but fail to stabilize
- channels that flicker and collapse
- boundaries and horizons that appear transiently
- curvature that oscillates around determinant-zero surfaces

This regime is universal because it arises whenever noise approaches the critical threshold or mobility falls below the coherence threshold. It is the architectural "edge of failure," where the motifs are visible in outline but cannot persist.

6.4 Non-Individuating Regime: Collapse of Architecture

The non-individuating regime appears whenever noise exceeds the critical threshold or mobility is too weak to amplify curvature. Across all conditions, this regime exhibits:

- no persistent saddles
- no coherent channels
- no boundaries or horizons
- churn dominated by stochastic fluctuations

The collapse of architecture in this regime is universal: it occurs in every parameter sweep, every initial condition, and every dimension. This demonstrates that the non-individuating regime is not a special case but a fundamental phase of the ED system.

6.5 Invariant Ordering of Regime Transitions

Across all parameter families, initial conditions, and dimensions, the regime transitions occur in the same order:

hyper-coherent → stable-coherent → marginal → non-individuating

This ordering is invariant under:

- mobility exponent variation
- noise amplitude variation
- suppression depth variation
- gradient amplification variation
- dimensional uplift

The universality of this ordering shows that the ED phase diagram is not a property of specific simulations but a structural consequence of the ED rule.

6.6 Regimes as Global Architectural Invariants

The universality of the regime structure demonstrates that ED possesses a global architectural identity. The same four regimes appear across all tested conditions, and the transitions between them follow the same sequence. This invariance is the macroscopic expression of the microscopic invariants—saddles, channels, boundaries, horizons, and curvature transitions—that define the ED ontology.

Regimes are therefore not empirical categories but architectural phases: necessary, reproducible, and universal.

7. Universality of Architectural Invariants

The preceding sections established that the motifs of Event Density—saddles, channels, boundaries, horizons, and regimes—persist under variation of parameters, initial conditions, physical measures, and dimensionality. In this section, we synthesize these results into a single claim: the architectural invariants of ED are universal. They arise not because of specific parameter choices, initial configurations, or dimensional constraints, but because they are the necessary geometric consequences of the ED rule. Universality is therefore not an empirical observation but an ontological statement: the ED architecture is intrinsic to the dynamics.

Architectural invariants are the structural features that persist under deformation. In ED, these invariants emerge from the interplay of gradient sign-flip, curvature transitions, mobility-driven sharpening, and noise-driven

perturbation. Each invariant corresponds to a geometric or dynamical necessity of the ED update. The universality of these invariants shows that ED is not a model with tunable behavior but a framework with unavoidable structure.

7.1 Saddle Formation from Gradient Sign-Flip

Across all conditions, saddles arise wherever the gradient reverses sign across orthogonal directions. This sign-flip condition is purely geometric and does not depend on mobility, noise, suppression depth, or dimensionality. Saddles are therefore universal because gradient sign-flip is universal. They are the minimal geometric units of ED architecture, and their persistence across all tests confirms their status as architectural invariants.

7.2 Channel Alignment from Coherent Gradients

Channels form wherever gradients maintain coherent direction along mixed-curvature corridors. This alignment condition arises from the ED update itself, which amplifies coherent gradients and suppresses incoherent fluctuations. Channels persist across parameter families, initial conditions, and dimensions because the mechanisms that generate them—gradient alignment and curvature guidance—are intrinsic to the rule. Channels are therefore universal conduits of individuating flow.

7.3 Boundaries and Horizons from Curvature Transitions

Boundaries and horizons arise from curvature transitions marked by determinant-zero surfaces of the Hessian. Positive curvature produces boundaries; negative curvature produces horizons; mixed curvature produces channels. These transitions occur in every tested configuration because curvature is an unavoidable consequence of the ED update. Boundaries and horizons are therefore universal surfaces that regulate compression and expansion in the field.

7.4 Regime Transitions Driven by Noise and Mobility

The four canonical regimes—hyper-coherent, stable-coherent, marginal, and non-individuating—appear in every parameter sweep and initial condition. Their transitions occur in the same order because they are driven by universal mechanisms: noise amplitude, mobility exponent, and curvature stability. The regime structure is therefore not a contingent classification but a universal phase diagram of the ED system.

7.5 Persistence Under Parameter Deformation

The architectural invariants persist across wide ranges of mobility exponent, noise amplitude, suppression depth, and nonlinear amplification. Parameter variation sharpens or softens motifs but does not eliminate them. This persistence shows that the invariants are not tied to specific parameter values but arise from the structural logic of the ED rule.

7.6 Independence from Initial Conditions

The invariants appear regardless of whether the system begins with random noise, smooth fields, structured patterns, or adversarial configurations. Initial conditions influence the transient dynamics but not the long-term architecture. This independence demonstrates that the invariants are not inherited from the initial state but generated by the ED update.

7.7 Dimensional Generalization

The invariants persist under dimensional uplift: saddles become saddle-sheets, channels become tubes, boundaries become walls, and horizons become surfaces. The same curvature transitions and gradient alignments appear in

3D as in 2D. This dimensional generalization shows that the invariants are not artifacts of planar geometry but expressions of deeper structural principles.

7.8 Synthesis: The Architectural Identity of ED

Across all axes of variation, the same conclusion emerges: the architectural invariants of Event Density are universal. They arise from the intrinsic geometry of the ED rule and persist under deformation of parameters, initial conditions, physical measures, and dimensionality. This universality defines the architectural identity of ED:

- Saddles anchor the architecture.
- Channels transmit individuating flow.
- Boundaries confine structure.
- Horizons regulate expansion.
- Regimes organize global behavior.

These motifs are not optional features; they are the necessary consequences of the ED rule. Universality elevates ED from a model to a framework, from a simulation to an ontology.

8. ED as a General Theory of Architectural Emergence

The universality results presented in this paper elevate Event Density from a specific dynamical rule to a general architectural framework. The persistence of saddles, channels, boundaries, horizons, and regime transitions across parameters, initial conditions, physical measures, and dimensions shows that these motifs are not contingent features of a particular implementation. They are the structural consequences of the ED rule itself. This section synthesizes these findings into a broader claim: ED provides a general theory of architectural emergence, in which coherent structure arises inevitably from the interplay of gradients, curvature, mobility, and noise.

Architectural emergence differs from traditional notions of pattern formation or dynamical stability. In ED, structure is not imposed by external constraints, boundary conditions, or finely tuned parameters. Instead, structure arises from intrinsic geometric mechanisms: gradient sign-flip generates saddles; curvature transitions generate boundaries and horizons; coherent gradient alignment generates channels; and noise and mobility regulate the stability of these motifs. These mechanisms operate universally, producing the same architectural motifs across scales, dimensions, and conditions. ED therefore offers a structural ontology: a minimal set of objects and relations that describe how individuating structure emerges in any relational field.

8.1 Architecture as Ontology

The ED ontology—saddles, channels, boundaries, horizons, and regimes—defines what exists in the system. These objects are not optional; they are necessary consequences of the ED rule. Universality shows that this ontology is not tied to specific parameter choices or initial conditions. It is the intrinsic structure of the ED field. The ontology is therefore not a descriptive convenience but a fundamental statement about the kinds of objects that can arise in ED dynamics.

8.2 Geometry as Mechanism

The mathematical structure of ED—gradient sign-flip, curvature transitions, determinant-zero surfaces, and mobility-driven sharpening—explains why the ontology must exist. These geometric mechanisms are dimension-agnostic and parameter-agnostic. They guarantee the formation of saddles, channels, boundaries, and

horizons in any sufficiently varied field. Geometry is therefore not a secondary description but the engine of emergence.

8.3 Regimes as Global Phases

The regime taxonomy—hyper-coherent, stable-coherent, marginal, and non-individuating—captures the global behavior of the ED field. Universality shows that these regimes are not empirical categories but architectural phases. They arise from the same geometric mechanisms that generate the local motifs. The regime structure is therefore the macroscopic expression of the ED ontology.

8.4 Universality as Evidence of Framework Status

The persistence of ED's architectural motifs across parameters, initial conditions, physical measures, and dimensions demonstrates that ED is not a model but a framework. Models require tuning; frameworks generate structure intrinsically. The universality of ED's motifs shows that the architecture is not contingent but necessary. This necessity is the defining feature of a scientific ontology.

8.5 Dimensional Scalability and Structural Depth

The natural generalization of ED's motifs from 2D to 3D—saddles to saddle-sheets, channels to tubes, boundaries to walls, horizons to surfaces—shows that ED's architecture scales with dimension. This scalability indicates that the ED ontology reflects deep structural principles rather than planar artifacts. Dimensional uplift therefore provides strong evidence that ED captures a general mechanism of architectural emergence.

8.6 ED as a Unified Architectural-Physical Framework

The alignment between the architectural layer (saddles, channels, boundaries, horizons) and the physical layer (energy, entropy, flux) shows that ED is a unified framework. The physical quantities introduced in ED-Arch-03 reinforce the architectural invariants, revealing their physical meaning. This convergence of geometry and physics is characteristic of mature scientific frameworks.

8.7 Toward a General Theory of Relational Structure

The universality of ED's motifs suggests that the ED rule captures a general mechanism by which structure emerges in relational systems. Whether the field represents density, probability, activation, or any other scalar quantity, the same architectural motifs arise. ED therefore provides a general theory of architectural emergence: a framework in which structure is not imposed but arises inevitably from the dynamics of the field.

9. Conclusion

The results presented in this paper demonstrate that the architectural motifs of Event Density—saddles, channels, boundaries, horizons, and coherent regimes—are universal consequences of the ED rule. They persist under variation of parameters, initial conditions, physical measures, and dimensionality. This universality shows that the architecture observed in ED-Arch-01, formalized in ED-Arch-02, and physically interpreted in ED-Arch-03 is not contingent on specific choices or configurations. It is intrinsic to the dynamics of the event density field.

Across all parameter families, the same motifs appear: gradient sign-flip generates saddles; curvature transitions generate boundaries and horizons; coherent gradient alignment generates channels; and noise and mobility regulate the stability of these structures. Across all initial conditions—random, smooth, structured, or adversarial—the system converges toward the same architectural attractors. Across all dimensions, the motifs generalize naturally: saddles become saddle-sheets, channels become tubes, boundaries become walls, and

horizons become surfaces. Across all physical measures, the same patterns of energy concentration, flux, and entropy reinforce the architectural invariants.

Taken together, these results establish ED as a general theory of architectural emergence. The ED ontology defines the fundamental objects of the system; the mathematical structure explains why they must exist; the regime taxonomy describes how they behave; and the universality results show that they cannot be avoided. The architecture is not imposed by initial conditions, parameter tuning, or dimensional constraints. It arises inevitably from the interplay of gradients, curvature, mobility, and noise.

The universality of ED's motifs positions the framework as a coherent architectural worldview: a system in which structure is not prescribed but emerges from the dynamics themselves. This worldview provides a foundation for future work, including the exploration of ED's cosmological implications, the development of higher-dimensional simulation engines, and the integration of ED with broader theories of relational structure. The architectural invariants identified here form the backbone of that program. They are the signature of ED's identity and the basis for its extension into new domains.